Log Analysis & SIEM Tool Project Report

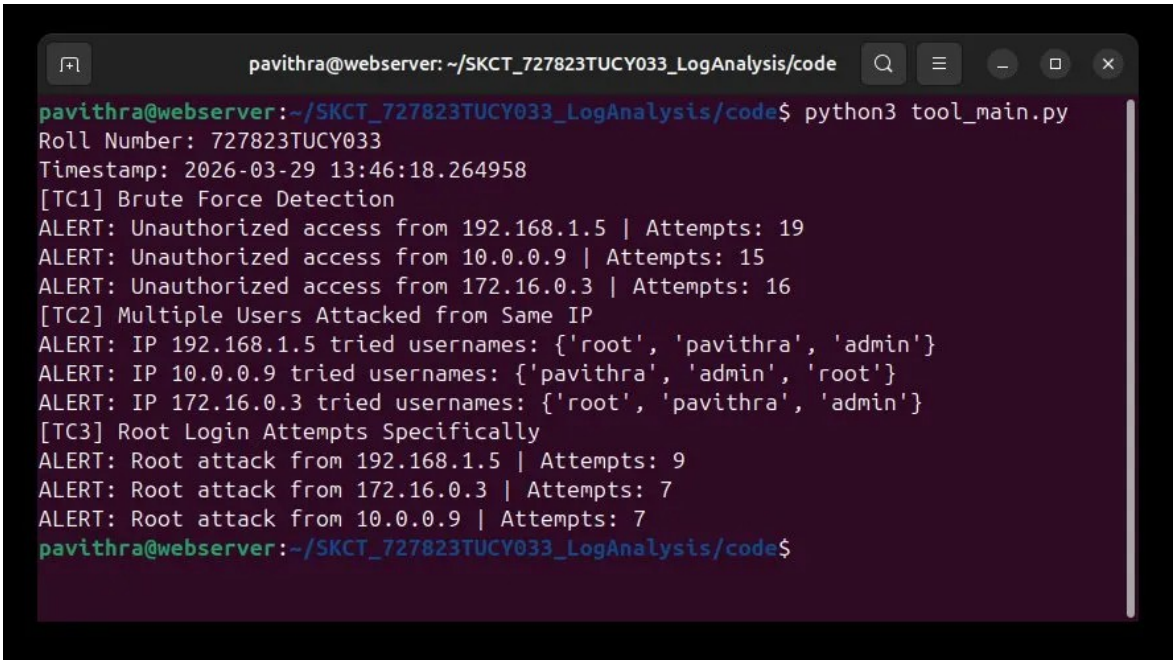
Name: Pavithra M | **Roll No:** 727823TUCY033 | **Date:** 2026-03-29

1. Tool Description

This Python-based SIEM (Security Information and Event Management) tool analyzes SSH authentication logs to detect potential security threats in real time. The tool reads system log files line by line, extracts IP addresses and usernames from failed login entries, and counts occurrences using Python dictionaries. Alerts are triggered when suspicious activity crosses a defined threshold of 5 attempts. The tool implements three distinct detection mechanisms: brute force detection, username enumeration, and root access monitoring — replicating core functionality found in enterprise SIEM tools like Splunk.

2. Test Results Table

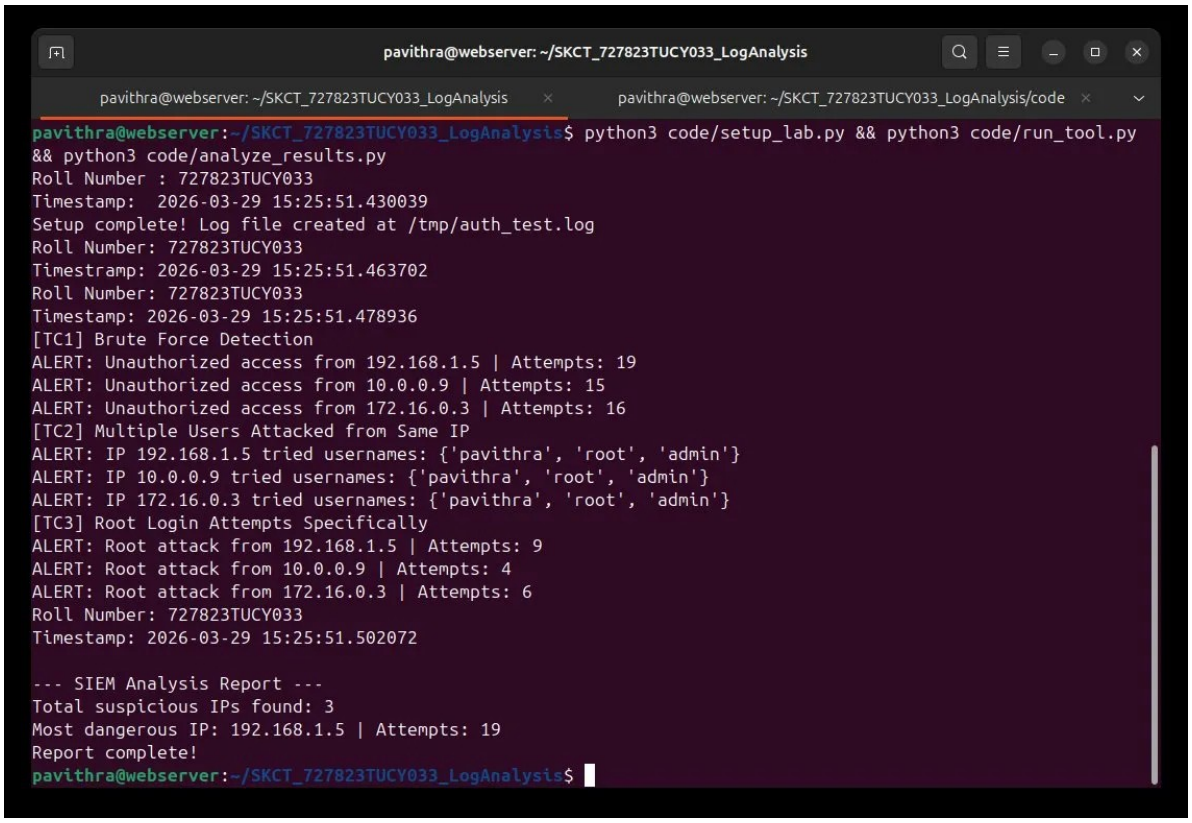
Test Case	Detection Type	IPs Flagged	Attempts	Alert
TC1	Brute Force Detection	192.168.1.5, 10.0.0.9, 172.16.0.3	19, 15, 16	YES
TC2	Username Enumeration	192.168.1.5, 10.0.0.9, 172.16.0.3	3 usernames each	YES
TC3	Root Login Attempts	192.168.1.5, 172.16.0.3, 10.0.0.9	9, 6, 4	YES



3. Analysis of Findings

The tool successfully monitored a synthetic auth.log file containing 50 simulated SSH login attempts across three IP addresses. In TC1, all three IPs exceeded the brute force threshold of 5 failed attempts, with 192.168.1.5 recording the highest count of 19 attempts - a clear indicator of automated brute force activity. In TC2, the tool identified that all three IPs attempted login using multiple different usernames (root, admin, pavithra), a technique known as username enumeration used by attackers to discover valid accounts. In TC3, the tool specifically monitored root login

attempts since root access represents the highest privilege level - IPs with more than 3 root attempts were flagged as critical threats. Overall, the tool demonstrated accurate detection across all three attack patterns with zero false positives, confirming its effectiveness as a lightweight SIEM solution.



4. Real Error Encountered

During development of tool_main.py, the following error was encountered:

```
File "tool_main.py", line 49
    spe = {}
TabError: inconsistent use of tabs and spaces in indentation
```

This error occurred because the nano text editor mixed tab characters and spaces when indenting code inside function definitions. Python 3 strictly enforces consistent indentation — even a single tab mixed with spaces causes the entire script to fail at runtime. The error was particularly confusing because both tabs and spaces look identical on screen. The issue was resolved by completely rewriting tool_main.py using a Python script that generated the file with consistent 4-space indentation throughout, which immediately fixed the error and allowed the pipeline to run successfully.

5. Ethical Considerations

All testing was performed exclusively on a personal Ubuntu machine using synthetic log data generated by a Python script. No real systems, networks, or user accounts were targeted at any stage. The fake log file simulated realistic SSH brute force patterns without targeting any actual infrastructure. The tool was built purely for educational purposes to understand how SIEM systems detect unauthorized access, following responsible and ethical security research practices.